

UNIT V
INTERFACES, THREADS AND GUI

Course Objective – Cob5: To learn interface and thread concept. Also, to learn the design of an GUI based application using JavaFX.

Course Outcome - CO5 : Design an interface, synchronize various tasks and develop an GUI using JavaFX.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1.	Define Interface. An Interface is basically a kind of class. Like classes, interfaces contain methods and variables with difference. That is interface do not specify any code to implement these methods. An interface in Java is a blueprint of a class. It has static constants and abstract methods. The interface in Java is a mechanism to achieve abstraction. There can be only abstract methods in the Java interface, not method body. It is used to achieve abstraction and multiple inheritance in Java.	K1
2.	How to implement an interface? The implements keyword is used to implement an interface. The interface keyword is used to declare a special type of class that only contains abstract methods. To access the interface methods, the interface must be "implemented" by another class with the implements keyword (instead of extends). The body of the interface method is provided by the "implement" class.	K1
3.	Define default interface. Default methods allow you to add new methods to interfaces without breaking the existing implementation of classes that already implement the interface. The default keyword in an interface method, which provides a default implementation. It has a body (implementation) unlike	K1

	traditional abstract methods in interfaces. Implementing classes can either use the default implementation or override it.	
4.	What is the thread? A thread is a lightweight subprocess. It is a separate path of execution because each thread runs in a different stack frame. A process may contain multiple threads. Threads share the process resources, but still, they execute independently.	K1
5.	How to create a thread in Java? There are two ways to create a thread: 1. By extending Thread class 2. By implementing Runnable interface. Thread class: Thread class provide constructors and methods to create and perform operations on a thread. Runnable interface: The Runnable interface should be implemented by any class whose instances are intended to be executed by a thread.	K1
6.	What is synchronization? Synchronization in java is the capability to control the access of multiple threads to any shared resource. In the Multithreading concept, multiple threads try to access the shared resources at a time to produce inconsistent results. The synchronization is necessary for reliable communication between threads.	K1
7.	What are the types of synchronization? Synchronization is classified into two types • Process Synchronization • Thread Synchronization 1. Process Synchronization in Java Process Synchronization is a technique used to coordinate the execution of multiple processes. It ensures that the shared resources are safe and in order. 2. Thread Synchronization in Java Thread Synchronization is used to coordinate and ordering of the execution of the threads in a multi-threaded program.	K1
8.	Define Inter – Thread Communication. It is a communication of two or more threads with each other. It can be done by using the following methods. • wait() • notify() • notifyAll()	K1
9.	Why we need Inter – Thread Communication?	K1

	* There is a situation on the thread that keeps on checking some conditions repeatedly, once that condition satisfies thread moves with the appropriate action. This situation is known as polling. This is a wastage of CPU time, to reduce the wastage of CPU time due to polling, java uses Inter – Thread Communication Mechanism. * wait(), notify(), notifyAll() methods must be called within a synchronized method or block otherwise program will compile but when you run it, it will throw illegal monitor State Exception.	
10.	Define wait() method. The wait() method causes current thread to release the lock and wait until either another thread invokes the notify() method or the notifyAll() method for this object, or a specified amount of time has elapsed.	K1
11.	Define notify() and notifyAll() methods. The notify() method wakes up a single thread that is waiting on this object's monitor. If any threads are waiting on this object, one of them is chosen to be awakened. The choice is arbitrary and occurs at the discretion of the implementation. The notifyAll() method wakes up all threads that are waiting on this object's monitor.	K1
12.	What about the daemon threads? The daemon threads are the low priority threads that provide the background support and services to the user threads. Daemon thread gets automatically terminated by the JVM if the program remains with the daemon thread only, and all other user threads are ended/died. There are two methods for daemon thread. - o public void setDaemon(boolean status): It used to mark the thread daemon thread or a user thread. - o public boolean isDaemon(): It checks the thread is daemon or not.	K1
13.	Define Multithreading.	K1

	Multitasking means executing several programs at a time, in system terminology, it is called multithreading. Thread is single line of execution with in the program.	
14.	Draw the Life Cycle of a Thread in Java Multithreading. <pre> graph TD NT[New Thread()] --> New New -- Start() --> Runnable Runnable -- run() --> Running Running -- Sleep(), wait() --> Waiting Waiting -- End of execution --> Dead Running -- End of execution --> Dead </pre>	K1
15.	What is Graphical User Interface (GUI)? A system of interactive visual components for a computer or system software is called a GUI (graphical user interface). GUI is the interface that uses graphical elements to let people interact as per requirement with electronic devices including computers, laptops, tablets, and smartphones. It is a very important component of software application programming since it substitutes actions for the text-based commands in the system. Whether it's a text file, object, image, or video as per requirement, it displays all types of required content a user could envision in the system.	K1
16.	What is JavaFX? JavaFX is a set of graphics and media packages that enables developers to design, create, test, debug and deploy rich client applications that operate consistently across different platforms. JavaFX makes it easier to create desktop applications and games in Java.	K1
17.	What are the key features of JavaFX? ● FXML and Scene Builder. ● Swing interoperability. ● Built-in UI controls and CSS. ● Canvas API. ● Multitouch Support ● Hardware-accelerated graphics pipeline. ● High-performance media engine. ● Self-contained application deployment model.	K1

18.	What are JavaFX Packages? The JavaFX elements are contained in packages that begin with the javafx prefix. At the time of this writing, there are more than 30 JavaFX packages in its API library. Here are four examples: javafx.application, javafx.stage, javafx.scene and javafx.scene.layout.	K1
19.	Define JavaFX controls. JavaFX controls are JavaFX components which provide some kind of control functionality inside a JavaFX application. Some of the controls are, ● Button ● CheckBox ● ColorPicker ● Label ● ListView ● Menubar ● RadioButton ● TextField ● ToggleButton	K1
20.	Define JavaFX Application Development. JavaFX application development refers to the process of creating desktop applications using the JavaFX framework, a set of Java libraries designed for building modern, feature-rich graphical user interfaces (GUIs) with advanced graphics, media, and web content. JavaFX provides a platform for developers to create applications that run on desktop systems (such as Windows, macOS, and Linux) and embedded devices.	K1

Part B

S.No.	Question	BTL *
1.	Explain briefly about Interfaces with an example.	K2
2.	How will you create a thread and explain with an example?	K2
3.	Explain Inter-Thread Communication with an example.	K2
4.	How will you synchronize a thread in Java. Explain with an example.	
5.	Explain about Multithreading with an example.	K2
6.	Explain about JavaFX controls.	K2
7.	Explain the purpose of using the TimeLine class and KeyFrame object in JavaFX.	K2

